

International Journal of Remote Sensing

ISSN: 0143-1161 (Print) 1366-5901 (Online) Journal homepage: https://www.tandfonline.com/loi/tres20

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**To cite this article:** Chunhua Liao, Jinfei Wang, Jiali Shang, Xiaodong Huang, Jiangui Liu & Ted Huffman (2018) Sensitivity study of Radarsat-2 polarimetric SAR to crop height and fractional vegetation cover of corn and wheat, International Journal of Remote Sensing, 39:5, 1475-1490, DOI: 10.1080/01431161.2017.1407046

**To link to this article:** https://doi.org/10.1080/01431161.2017.1407046

Published online: 29 Nov 2017.

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INTERNATIONAL JOURNAL OF REMOTE SENSING, 2018
VOL. 39, NO. 5, 1475–1490
https://doi.org/10.1080/01431161.2017.1407046

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# Sensitivity study of Radarsat-2 polarimetric SAR to crop height and fractional vegetation cover of corn and wheat

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**ABSTRACT**

Increasing studies have been conducted to investigate the potential of polarimetric synthetic aperture radar (SAR) in crop growth monitoring due to the capability of penetrating the clouds, haze, light rain, and vegetation canopy. This study investigated the sensitivity of 16 parameters derived from C-band Radarsat-2 polarimetric SAR data to crop height and fractional vegetation cover (FVC) of corn and wheat. The in-situ measured crop height and FVC were collected from 29 April to 30 September 2013, at the study site in southwest Ontario, Canada. A total of 10 Radarsat-2 polarimetric SAR images were acquired throughout the same growing season. It was observed that at the early growing stage, the corn height was strongly correlated with the SAR parameters including HV ($R^2$ = 0.88), HH-VV ($R^2$ = 0.84), and HV/VV ($R^2$ = 0.80), and the corn FVC was significantly correlated with HV ($R^2$ = 0.79) and HV/VV ($R^2$ = 0.92), but the correlation became weaker at the later growing stage. The sensitivity of the SAR parameters to wheat variables was very low and only HV and Yamaguchi helix scattering showed relatively good but negative correlations with wheat height ($R^2$ = 0.57 and $R^2$ = 0.39) at the middle growing stage. These findings indicated that Radarsat-2 polarimetric SAR (C-band) has a great potential in crop height and FVC estimation for broad-leaf crops, as well as identifying the changes in crop canopy structures and phenology.

**ARTICLE HISTORY**
Received 26 August 2016
Accepted 11 November 2017

## 1. Introduction

Corn and wheat are two different types of crops according to their leaf size (broad-leaf crop *vs* narrow-leaf crop). They are two of the valuable crops in Canada and play an important role in the global agricultural trading market (Farm Credit Canada 2013; Hamel and Dorff 2014). The corn and wheat planted in Canada are not only for grain but also for silage. Therefore, crop growth monitoring enables the farmers to make timely decisions on crop management in order to maximize crop production. Crop height is closely related to crop biomass and phenology and is an important indicator for crop growth monitoring, crop discrimination, and crop production estimation (McNairn and Brisco 2004; Srivastava, Patel, and Navalgund 2006). The fractional vegetation cover (FVC) is defined as the fraction of

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green vegetation area including leaves and stems projected on the flat surface in unit area. FVC is an important structural property of a plant canopy, and a key variable for describing the vegetation coverage. It was found that FVC has a linear relationship with the fraction of absorbed photosynthetically active radiation (FAPAR) of crop canopies, which is an important variable for crop biomass estimation (Xie et al. 2013). Compared with leaf area index (LAI), both the LAI and FVC can be calculated from the gap fractions derived from the digital hemispherical photographs (Mougin et al. 2014), whereas the uncertainties of FVC are lower than the effective LAI (Verger et al. 2009).

The use of Earth Observation technology has provided an efficient way for various agricultural applications such as crop variable estimation, crop type inventory, and crop yield prediction due to its capability of providing timely and large spatial coverage of land surface information over a wide range of spatial and temporal scales. However, optical image acquisition relies heavily on the weather condition, and the crop variable estimation over regions with frequent cloud cover usually cannot meet the temporal requirement for agricultural applications during the growing season. Besides, the estimation of crop structure-related variables such as crop height is difficult using optical remote sensing (Srivastava, Patel, and Navalgund 2006). The multi-temporal polarimetric synthetic aperture radar (SAR) data can be used to monitor the temporal variations of crop growth more continuously than optical data due to its capability of penetrating the clouds, haze, light rain, and vegetation canopy. Polarimetric SAR data have great potential in estimating crop variables because SAR parameters are sensitive to many vegetation biophysical variables such as plant structure, leaf size, stem density, biomass, and plant water content (McNairn and Brisco 2004; Srivastava, Patel, and Navalgund 2006). However, they are also sensitive to the parameters of the underlying soil such as soil moisture and surface roughness (McNairn and Brisco 2004; Jiao et al. 2011), and the responses of SAR backscatter to crop biophysical characteristics vary with SAR frequency, incidence angle, and polarizations (Jiao et al. 2011), which makes the application of SAR in agriculture more complicated. Previous research has revealed that polarimetric SAR parameters are sensitive to crop LAI (Jiao et al. 2011) and crop biomass (Mattia et al. 2003; Wiseman et al. 2014) which is related to crop phenological development (Shang et al. 2013). Jiao et al. (2010) investigated the sensitivity of X-band, C-band, and L-band polarimetric SAR backscatter signatures to corn and soybean LAI. High correlations were observed between L-band and C-band polarimetric SAR backscatter and corn and soybean LAI, whereas X-band backscatter was poorly correlated with both corn and soybean LAI. C-band SAR data have relatively longer wavelength than X-band SAR data but shorter wavelength than L-band SAR data, and thus they can penetrate into the crop canopy but are less affected by soil due to reduced penetration into the deep crop canopy (McNairn and Brisco 2004). Therefore, the C-band polarimetric SAR data was regarded as a good candidate for biomass and LAI estimation of crops (Ferrazzoli et al. 1997; Lin et al. 2009).

Radarsat-2 is a C-band (5.3 GHz) polarimetric SAR system with the spatial resolutions varying from 3 to 100 m. Although the orbit repeat cycle is 24 days, the flexibility of the steerable radar beam makes the revisit intervals shorter. With the available of quad-polarization data from satellites such as Radarsat-2, it is possible to study the sensitivity of more polarimetric SAR parameters including the four polarizations (HH, HV, VH, and VV) and several decompositions extracted from the four polarizations scattering matrix. Previous studies have investigated the sensitivity of SAR parameters to crop biophysical variables such as LAI and biomass, and it was observed that the responses of SAR backscatter to LAI or

biomass of narrow-leaf crops such as wheat were different from the responses of SAR backscatter to the LAI or biomass of broad-leaf crops such as corn (Macelloni et al. 2001; Mattia et al. 2003; Smith et al. 2006; Fontanelli et al. 2013; Wiseman et al. 2014). Besides LAI and biomass, crop height is an important crop variable in vegetation growth dynamics monitoring, and FVC has been used for crop biomass estimation and crop evapotranspiration modelling (Paruelo, Lauenroth, and Roset 2000; Singh, Dutta, and Dharaiya 2013). If these crop variables can be estimated from Radarsat-2 polarimetric SAR data, the high temporal frequency requirement can be met. However, the responses of Radarsat-2 polarimetric SAR backscatter to crop height and FVC were not well documented in the literature. In order to investigate the potential of Radarsat-2 polarimetric SAR in crop height and crop FVC estimation and monitoring, the objectives of this study are (1) to investigate the sensitivity of different Radarsat-2 polarimetric SAR parameters to crop height and FVC of corn and wheat and (2) to explore the variations in SAR responses to crop height and FVC caused by the differences in crop growth stage and canopy structure.

## 2. Methodology

### 2.1. *Study site*

The study site was selected near Stratford, Ontario, Canada (43.3° N, 80.8° W, Figure 1), a place located in the Mixedwood Plains Ecozone, and characterized by abundant water

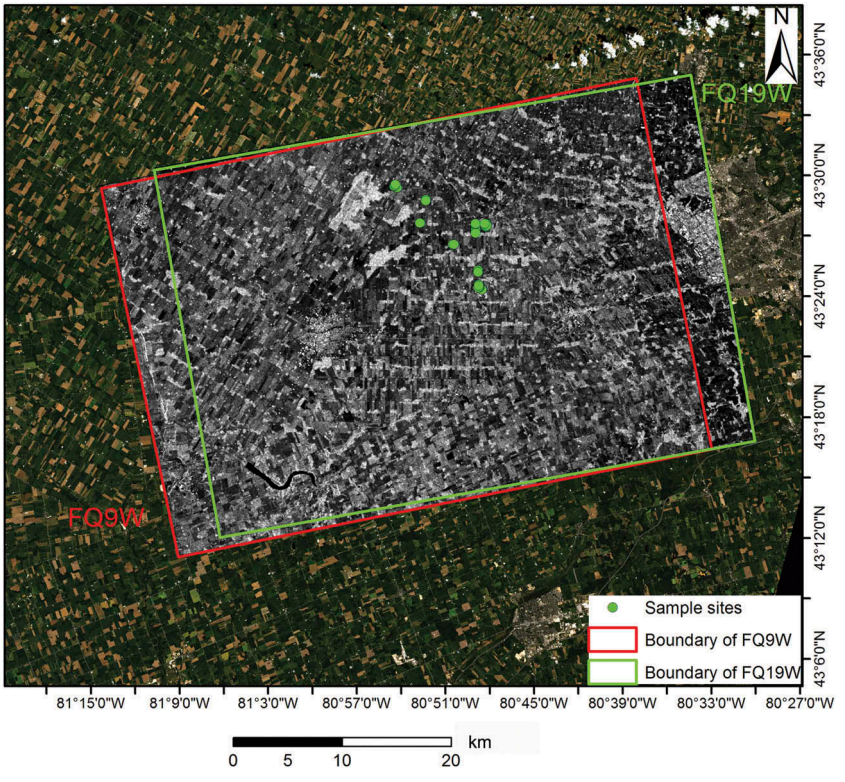



Figure 1. Map of study site and the sample sites.

supply, ideal weather condition, and productive soil for agriculture. The study site's average elevation is about 350 m above sea level with a relatively flat terrain. Corn and winter wheat are two major crops grown in this area. Corn in our study area is generally seeded in May and harvested in September or October, and winter wheat is seeded in previous October and harvested in July. Five corn fields and four winter wheat fields were selected for intensive field measurements and sampling throughout the growing season in 2013. The fields were within a 100 $km^2$ area.

## 2.2. *Field data collection and analysis*

A total of 17 sample sites located in five corn fields and 18 sample sites in four winter wheat fields were deployed. Field data collected in each sample site include (1) taking 14 hemispherical photographs for each sampling site (seven photographs along one row and then another seven photographs along another row), (2) collecting crop height measurements and recording crop phenological stages using the Bundesanstalt, Bundessortenamt, and Chemical (BBCH) scale at each sample site. For corn fields, the photos were taken by the camera facing downward if the plant height of corn was less than 1.2 m, otherwise, the photos were taken upward. For wheat fields, the photos were all taken downwards. The FVC was extracted from the 14 hemispherical photographs using *CanEye*. Fieldwork was conducted from April to August 2013, concurrently with Radarsat-2 satellite passing over. Figure 2 gives a synopsis of the corn and winter wheat at different growing stages during the period under study.

Crop height was related to crop phenology and crop biomass, especially at the early growing stage. According to the phenological stage survey, the relationship between the BBCH scale and crop height is shown in Figure 3(a,b). In this study site, when the corn height was around 150 cm, the BBCH scale was observed as 51, and the phenology of corn reached to the beginning of heading and tasselling. When the wheat height was greater than 65 cm, the BBCH scale was observed as 59, the end of heading. It was observed that different wheat height could have the same wheat BBCH. This may be because that the difference of wheat varieties and growth conditions lead to the difference of wheat height even though they are at the same phenology. Experience in fields showed that plants canopy can have different heights or LAI values in the same field – with the same BBCH.

Corn FVC showed a strong correlation with corn height at the early growing stages but became less responsive when the height was greater than 150 cm, as the plants continued growth in the vertical dimension after canopy reached closure. For wheat, FVC also showed a saturation phenomenon when wheat height was greater than 45 cm, and it varied from nearly 60% to 85% (Figure 4). According to the definition of FVC, this phenomenon for wheat may be because the dense plants and narrow leaves of wheat make the stems contribute a large percentage to the FVC, and accordingly lead to the saturation of FVC at the early stage. In addition, this large variation of FVC at the later stage may be caused by the difference of wheat varieties.

## 2.3. *Radarsat-2 data acquisition and processing*

The characteristics of the Radarsat-2 data used in this study and the phenology of the winter wheat and corn associated with each image are shown in Table 1. Radarsat-2

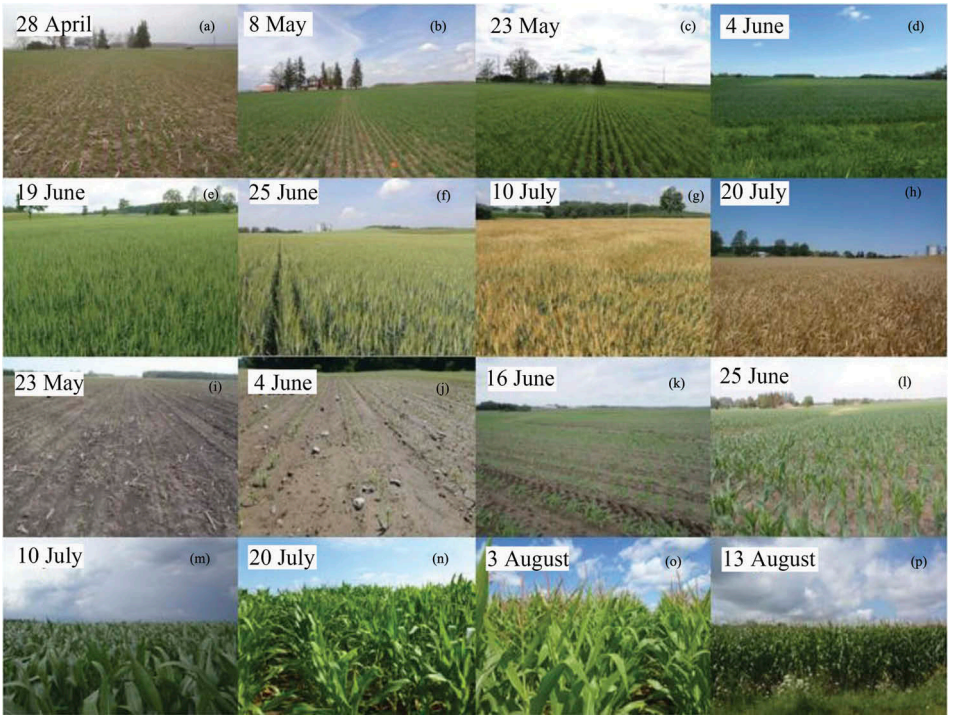



**Figure 2.** Wheat (a–h) and corn (i–p) growth stages of different periods: (a) leaf development, (b) tillering, (c) beginning of stem elongation, (d) middle of stem elongation, (e) end of heading, (f) development of fruit, (g) ripening (hard dough), (h) fully ripe; (i) leaf development (3 leaves), (j) leaf development (4 leaves), (k) leaf development (6 leaves), (l) beginning of stem elongation, (m) end of stem elongation, (n) tassel emergence; (o) tassel in flower and stigmata fully emerged, (p) end of flowering and stigmata completely dry.

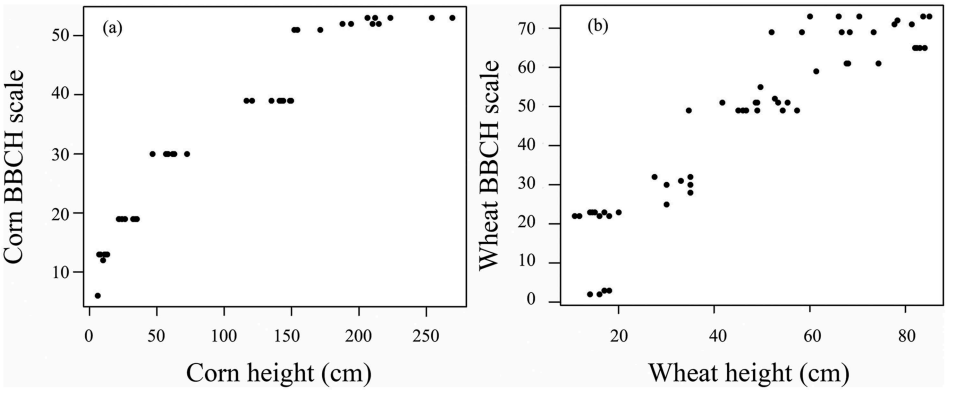



**Figure 3.** Relationship between crop height and BBCH scale for corn (a) and winter wheat (b).

polarimetric SAR can provide full polarimetric data with its Wide Fine Quad-polarization mode at different incidence angles. The temporal resolution for the same mode is 24 days. However, most of the crops such as corn in North America grow rapidly in the summer season. For example, during the 16-day period from 25 June to 11 July, the average height

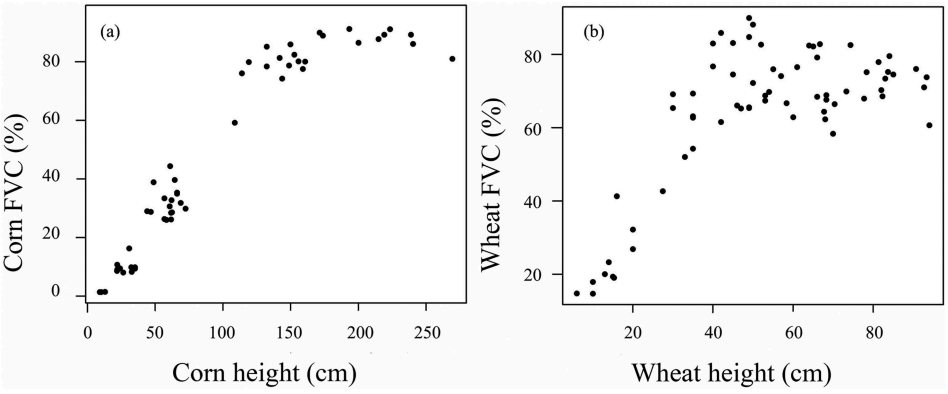



**Figure 4.** Correlation between FVC and corn height (a) and wheat height (b).

**Table 1.** The characteristics of the acquired Radarsat-2 data and the phenology of crops associated with each image.

| Date | Sensor type | Incidence angle (°) | Beam | Crop type | Growing stage of most crops |
|---|---|---|---|---|---|
| 29 April 2013 | Radarsat-2 | 27.2 | FQ9W | Wheat | Leaf development (5 leaves) |
| 8 May 2013 | Radarsat-2 | 37.7 | FQ19W | Wheat | Beginning of tillering |
| 23 May 2013 | Radarsat-2 | 27.2 | FQ9W | Wheat/corn | Beginning of stem elongation/leaf development (3 leaves) |
| 2 June 2013 | Radarsat-2 | 37.7 | FQ19W | Wheat/corn | Middle of stem elongation/leaf development (4 leaves) |
| 16 June 2013 | Radarsat-2 | 27.2 | FQ9W | Wheat/corn | End of heading/leaf development (6 leaves) |
| 26 June 2013 | Radarsat-2 | 37.7 | FQ19W | Wheat/corn | Development of fruit/beginning of stem elongation |
| 10 July 2013 | Radarsat-2 | 27.2 | FQ9W | Wheat/corn | Ripening (hard dough)/end of stem elongation |
| 20 July 2013 | Radarsat-2 | 37.7 | FQ19W | Wheat/corn | Fully ripe/Tassel emergence |
| 3 August 2013 | Radarsat-2 | 27.2 | FQ9W | Corn | Tassel in flower and stigmata fully emerged |
| 13 August 2013 | Radarsat-2 | 37.7 | FQ19W | Corn | End of flowering and stigmata completely dry |

of corn changed from 60 cm to 145 cm. So if only the polarimetric SAR images acquired at the same mode were used, there will be a lack of data at some important phonological stage. For this reason, 10 SAR images at two modes, FQ9W (incidence angle: 27.2°–30.5°) and FQ19W (incidence angle: 37.7°–40.4°) during the growing season from 29 April 2013 to 13 August 2013, were acquired and analysed together in this study, and the time intervals were shortened to 10–14 days. The difference of the two incidence angles is about 10°. Even though the backscatter will decrease slightly with the increase of incidence angle (McNairn et al. 2000), the sensitivity will not change significantly when the difference of the two incidence angles is less or equal to 10° if the data for each incidence angle could cover all phenological stages (Fontanelli et al. 2013).

Radarsat-2 data were processed using PolSARPro and the final spatial resolution of the Radarsat-2 data was 12.5 m. A 7 × 7 boxcar filter was applied to the data by averaging the incoherent noise caused by the random interference between SAR scatters (Goodman 1976) to suppress the speckle. Then, orthorectification was performed using *MapReady*. A variety of parameters can be extracted from the Radarsat-2 polarimetric SAR data. The covariance matrix ($\mathbf{C}_3$) and coherency matrix ($\mathbf{T}_3$) and are

convertible fundamental matrices, from which other decomposition parameters can be derived. The covariance matrix ($\mathbf{C}_3$) was extracted from the single look complex format SAR data, which can be described by a scattering matrix **S** (Equation (1)) (Lee and Pottier 2009a):

$$\mathbf{S} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \tag{1}$$

The four elements of the **S** matrix $S_{HH}$, $S_{HV}$, $S_{VH}$, and $S_{VV}$ represent the four polarizations, with each polarization having a complex pair.

The covariance matrix $\mathbf{C}_3$ can be expressed as Equation (2):

$$\begin{aligned} \mathbf{C}_3 = \mathbf{K}_C \cdot \mathbf{K}_C{}^{*T} &= \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \\ &= \begin{bmatrix} |S_{HH}|^2 & \sqrt{2}S_{HH}S_{HV}{}^* & S_{HH}S_{VV}{}^* \\ \sqrt{2}S_{hv}S_{HH}{}^* & 2|S_{HV}|^2 & \sqrt{2}S_{HV}S_{VV}{}^* \\ S_{VV}S_{HH}{}^* & 2S_{HV}(S_{HH} - S_{VV})^* & |S_{VV}|^2 \end{bmatrix} \end{aligned} \tag{2}$$

where $\mathbf{K_C} = [S_{HH}\ \sqrt{2}S_{HV}\ S_{VV}]^T$. For the monocratic case, $S_{HV} = S_{VH}$, $*$ denotes the conjugate and $|\ |$ denotes the modulus.

The coherency matrix $\mathbf{T}_3$ can be expressed as Equation (3):

$$\begin{aligned} \mathbf{T}_3 = \mathbf{K_T} \cdot \mathbf{K_T}{}^{*T} &= \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \\ &= \frac{1}{2}\begin{bmatrix} |S_{HH} + S_{VV}|^2 & (S_{HH} + S_{VV})(S_{HH} - S_{VV})^* & 2(S_{HH} + S_{VV})S_{HV}{}^* \\ (S_{HH} - S_{VV})(S_{HH} + S_{VV})^* & |S_{HH} - S_{VV}|^2 & 2(S_{HH} - S_{vv})S_{HV}{}^* \\ 2S_{HV}(S_{HH} + S_{VV})^* & 2S_{HV}(S_{HH} - S_{VV})^* & 4|S_{HV}|^2 \end{bmatrix} \end{aligned} \tag{3}$$

where $\mathbf{K_T} = \frac{1}{\sqrt{2}}[S_{HH} + S_{VV}S_{HH} - S_{VV}\ S_{HV} + S_{VH}]^T$.

In the covariance matrix, the upper or lower triangular elements are complex numbers. The diagonal elements $C_{11}$, $C_{22}$, and $C_{33}$ represent the backscattering coefficients of the targets in horizontal polarization (HH), cross polarization (HV), and vertical polarization (VV) respectively ($C_{11} = \sigma^0{}_{HH}$, $C_{22} = \sigma^0{}_{HV}$, $C_{33} = \sigma^0{}_{VV}$) (Duguay et al. 2015), and they are associated with the structural characteristics of the targets. The three diagonal elements in the coherency matrix ($\mathbf{T}_3$) matrix, $T_{11}$ (HH+VV), $T_{22}$ (HH-VV), and $T_{33}$ (VV) are Pauli decomposition parameters and are associated with single-bounce scattering, double-bounce scattering, and volume scattering respectively (Lee and Pottier 2009b). It has been reported that the element $T_{22}$ (HH-VV) of Radarsat-2 Pauli-basis decomposition is a good indicator of crop growth development (Liu et al. 2013; Wiseman et al. 2014). Because the element $T_{33}$ is the same as $C_{22}$, which represents HV polarization, $T_{33}$ is not analysed in this study.

The Cloude–Pottier decomposition, Yamaguchi 4-component decomposition, radar vegetation index (RVI), as well as the intensity ratio (HH/VV, HV/VV, HV/HH) were obtained from the covariance matrix. These polarimetric SAR parameters reflect the scattering characteristics associated with the structural information of targets, such as the dominant type of scattering, the randomness of the scattering. The Cloude–Pottier

decomposition includes entropy ($H$), anisotropy ($A$), and alpha angle ($\alpha$). Entropy ($H$) is the randomness of the scattering, which ranges between 0 (coherent) and 1 (pure random). The alpha angle identifies the type of scattering, which varies between the irregular surface (0°–45°) and double-bounce scattering (45°–90°). The anisotropy ($A$) represents the relative difference between the second and third eigenvalues (Cloude and Pottier 1996). The Yamaguchi 4-component decomposition is a modification of the Freeman–Durden decomposition. It decomposes covariance matrix into volume scattering, helix scattering, double-bounce scattering, and single-bounce scattering. The surface scattering and double-bounce components are the same as in Freeman–Durden decomposition. The helix component is added to interpret areas with sharp corners and edges over urban areas (Yamaguchi et al. 2005).

The RVI was proposed to monitor the vegetation growth condition (Kim and Van Zyl 2009), and it was found that RVI had a good correlation with different vegetation indices such as LAI and NDVI as well as the crop water content. The RVI was calculated by Equation (4).

$$\mathrm{RVI} = \frac{8\sigma^0{}_{\mathrm{HV}}}{\sigma^0{}_{\mathrm{HH}} + \sigma^0{}_{\mathrm{VV}} + 2\sigma^0{}_{\mathrm{HV}}} \tag{4}$$

where $\sigma^0{}_{\mathrm{HH}}$, $\sigma^0{}_{\mathrm{HV}}$, and $\sigma^0{}_{\mathrm{VV}}$ represent the backscatter at HH, HV, and VV polarization.

### 2.4. *Regression analysis*

Regression analysis between Radarsat-2 polarimetric SAR parameters and crop variables was conducted to evaluate the sensitivity of SAR parameters to crop variables through the course of the growth cycle. The crop height and FVC were collected through fieldwork from April to August. The scatterplots of SAR response to crop height and FVC show different patterns or trends at different crop stages. Therefore, the regression analysis between SAR parameters and crop variables was conducted separately at different growing stages of corn and wheat.

## 3. Results and discussion

The SAR data obtained at two different incidence angles (27.2° and 37.7°) were combined in this study in order to shorten the revisit time and capture the important phenological changes of crops during the growing season. From the scatterplots between different Radarsat-2 polarimetric SAR parameters, including linear backscatter coefficients, Pauli decomposition, Cloude–Pottier decomposition, Yamaguchi four-component decomposition parameters, radar vegetation index and the intensity ratios (HH/VV, HV/VV, HV/HH), and crop variables (crop height and FVC), it was found that the responses of SAR parameters to the crop variables were different for corn and wheat, and the responses were also different at different growing stages. Corn and wheat are two distinct types of crops with different leaf size, orientation, and canopy structures. Corn has a much broader leaf size than wheat, and the contribution of corn leaves to backscattering dominates and the contribution of stems is attenuated for the broadleaf crop (Macelloni et al. 2001; Paloscia et al. 2014). Wheat leaf is narrow, and the contribution of the leaves and stems to backscattering is comparable (Macelloni et al. 2001;

Paloscia et al. 2014). In addition, the soil below, the canopy, and the change of crop physical structure can also affect the SAR backscatter, and the penetrating capability of C-band SAR signal can be weakened by the growing canopy volume and accumulation of biomass.

### 3.1. *Correlation analysis between Radarsat-2 polarimetric SAR parameters and crop height*

From the scatterplots between corn height and different SAR parameters, it was found that the corn height was correlated with most SAR parameters, especially HV, HH-VV, and HV/VV, when the height was less than 150 cm or before the heading of corn (hereafter we call it the early stage). When the corn grew taller than 150 cm (hereafter we call it the later stage), the sensitivity of most SAR parameters to the corn height was lost. From the scatterplots between wheat height and different SAR parameters, three distinctive stages were observed for most SAR parameters. The two height thresholds for the three stages were 25 cm (middle of tillering) and 65 cm (end of heading) (hereafter we call the three stages the early stage, the middle stage, and the later stage).

Table 2 provides the coefficient of determination ($R^2$) between in-situ measured crop height and the Radarsat-2 polarimetric SAR parameters for corn and wheat at two different growing stages. The sensitivity of polarimetric SAR parameters to crop variables was higher for corn than for wheat, which is consistent with the finding by Wiseman et al. (2014) in their study over western Canada. It was also found that most polarimetric SAR parameters had negative relationship with wheat height, and the correlation was very weak. At the early stage when wheat height was less than 25 cm, SAR backscatter

**Table 2.** Coefficient of determination ($R^2$) between *in situ* measured crop height and Radarsat-2 polarimetric SAR parameters (the minus means negative correlation).

| | Corn | | Wheat | |
|---|---|---|---|---|
| SAR parameters | Height <150 cm | Height ≥ 150 cm | 25 ≤ Height <65 cm | Height ≥ 65 cm |
| Linear polarization | | | | |
| HH($C_{11}$) | 0.75* | **−0.27*** | −0.08 | −0.09 |
| HV($C_{22}$) | **0.88*** | −0.19 | **−0.57*** | −0.04 |
| VV($C_{33}$) | 0.28* | −0.02 | −0.20* | −0.14* |
| Pauli decomposition | | | | |
| HH+VV($T_{11}$) | 0.34* | −0.02 | −0.21* | **−0.15*** |
| HH-VV ($T_{22}$) | **0.84*** | **−0.59*** | −0.01 | −0.1* |
| Intensity ratio | | | | |
| HH/VV | 0.58* | −0.16 | 0.12 | 0.03 |
| HV/HH | 0.40* | 0.04 | −0.30* | 0.01 |
| HV/VV | **0.80*** | 0.01 | −0.02 | 0.13* |
| Radar vegetation index | | | | |
| RVI | 0.53* | 0.13 | −0.26* | 0.07 |
| Cloude-Pottier decomposition | | | | |
| Entropy($H$) | 0.47* | −0.05 | −0.21* | −0.01 |
| Anisotropy ($A$) | −0.28* | −0.19 | 0.10 | −0.01 |
| Alpha ($\alpha$) | −0.38* | 0.17 | −0.20* | −0.02 |
| Yamaguchi 4-component decomposition | | | | |
| Yamaguchi single bounce | 0.15* | 0 | −0.11 | −0.16 |
| Yamaguchi double bounce | 0.61* | −0.46* | 0.09 | 0 |
| Yamaguchi volume scattering | 0.64* | −0.25* | −0.04 | 0 |
| Yamaguchi helix scattering | 0.66* | −0.09 | −0.39* | −0.06 |

*$p$-Value <0.01.

was not responsive to wheat height due to the dominant influence of soil roughness and soil moisture. Therefore, the analysis at the early stage was not listed in Table 2.

Figure 5 illustrates the relationships between the corn height and HV (a) and HH-VV (b) for the two modes (FQ9W and FQ19W). The HV polarization, dominated by volume scattering, was significantly correlated with the corn height at the early stage ($R^2$ = 0.88). At the later stage, the HV was no longer sensitive to the corn height. This is because that HV is primarily representing the volume scattering at early growing stages, but once the corn grows taller and denser, the SAR penetration is limited and the HV will be reduced due to the attenuation effect. Therefore, L-band polarimetric SAR may be more useful for the corn height estimation for the later growing stage. The HH-VV is strongly correlated with the corn height at the early stage ($R^2$ = 0.84), and a good negative correlation was also observed at the later stage ($R^2$ = 0.59). This correlation was also supported by Liu's work addressing that HH-VV was strongly correlated with corn LAI, and HH-VV decreased as the LAI increased in the later stage (Liu et al. 2013). For the HH-VV, it actually consists of surface, double-bounce and volume scattering. At different stages of corn, the dominant scattering may be different. If there is only one dominant scattering, the correlation with the height would be high. Otherwise, the correlation would be lower. The good correlation between HH-VV and corn height at both stages, even though the trends of the correlations are opposite, indicated that the HH-VV has a great potential in corn height monitoring and estimation through the whole growing season.

Figure 2 shows that most SAR parameters except HV and Yamaguchi helix scattering were not sensitive to wheat height in this study site. Figure 6 illustrates that the relationship between the wheat height and HV (Figure 6a) and Yamaguchi helix scattering (Figure 6b) for the two modes (FQ9W and FQ19W). At the early stage when the wheat height was less than 25 cm, no SAR parameters were sensitive to wheat height. The HV backscatter from FQ9W (29 April) was much higher than that from FQ19W (8 May). This is perhaps due to the much higher soil moisture after the rain on 29 April when the FQ9W mode image was acquired (Baghdadi et al. 2007; Shoshany et al. 2000), the large soil roughness, or crop residues (Huang, Wang, and Shang 2016) at the early growing stage. It was also observed that wheat height was negatively correlated with HV

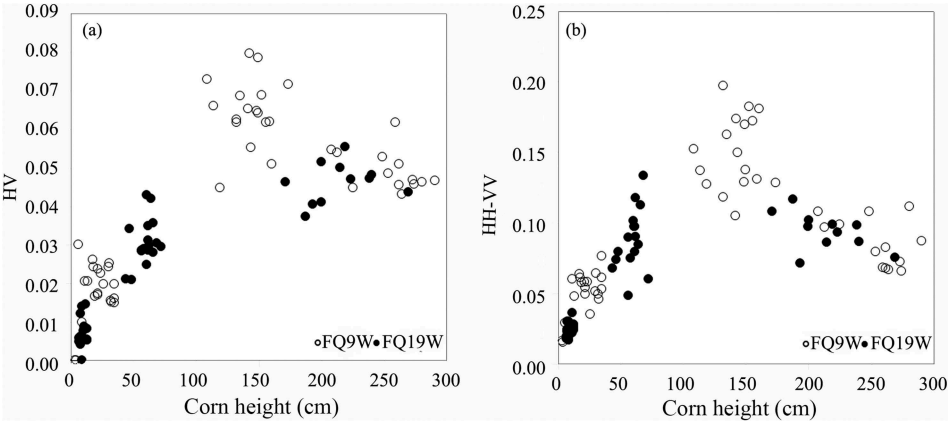



**Figure 5.** Correlation between corn height and HV ($C_{22}$) (a) and HH-VV ($T_{22}$) (b).

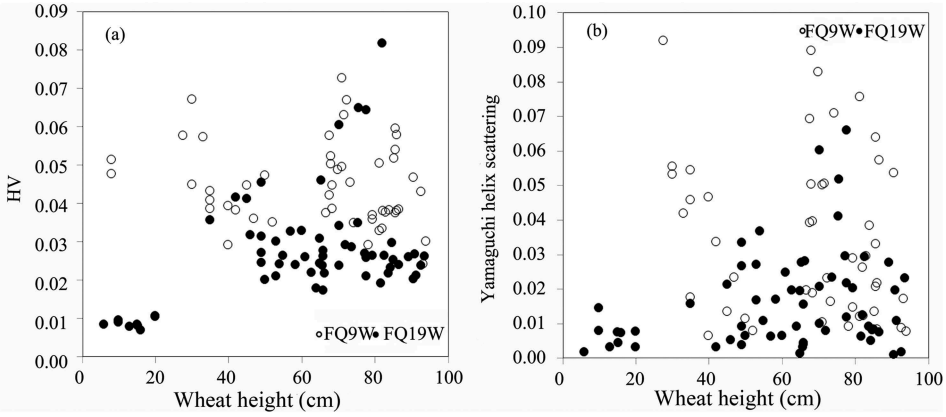



**Figure 6.** Correlation between wheat height and HV ($C_{22}$) (a) and Yamaguchi helix scattering (b).

($R^2$ = 0.57) and Yamaguchi helix scattering ($R^2$ = 0.39) at the middle stage. This may be because that the dense vertical and hallow stems would absorb the backscatter (Fontanelli et al. 2013; Macelloni et al. 2001), hence it results in the decrease of backscatter with the growth of wheat during the stem elongation. At the later stage, the wheat was in its heading and ripening stage, the HV and Yamaguchi helix scattering increased drastically and was no longer sensitive to wheat height. It is likely that the emergence of inflorescence led to the change of the wheat canopy and affected the responses of SAR parameters. In addition, as the wheat kernels developed, the inside of them changed from a liquid state to dough, and became denser in consistency and thus resulting in the increase in SAR backscatters (Wiseman et al. 2014). Due to the variations of HV and Yamaguchi helix scattering before and after the heading stage, they may be applied to the wheat phenology detection.

### 3.2. *Correlation analysis between Radarsat-2 polarimetric SAR parameters and FVC*

Table 3 shows the coefficient of determination ($R^2$) between the in-situ measured FVC and Radarsat-2 polarimetric SAR parameters for corn and wheat at two different growing stages. Previous research has found that when crop (corn and soybean) LAI reached to a certain value (LAI >3) (Jiao et al. 2011), the sensitivity of Radarsat-2 polarimetric SAR parameters to crop LAI decreased. Similar phenomenon also exists in the regression analysis between SAR parameters and corn FVC; all SAR parameters were insensitive to corn FVC when FVC was greater than 75% (the beginning of tassel emergence). For wheat FVC, because of the influence of soil at the early stage, and the saturation and the large variation of FVC at later stage (Figure 4), no SAR parameters were found sensitive to wheat FVC at any stage.

Among the three linear polarizations (HH, HV, VV), HV showed a good correlation with the corn FVC ($R^2$ = 0.79) when the FVC was below 75%. The VV barely showed any correlation with FVC at both stages. It suggests that the C-band VV polarization is not an effective parameter to monitor broadleaf plants. Similar results were also found in

**Table 3.** Coefficient of determination ($R^2$) between *in situ* measured FVC and Radarsat-2 polarimetric SAR parameters (the minus means negative correlation).

| | Corn | | Wheat | |
|---|---|---|---|---|
| SAR parameters | FVC < 75% | FVC ≥ 75% | FVC < 42% | FVC ≥ 42% |
| Linear polarization | | | | |
| HH($C_{11}$) | 0.35* | 0.3* | 0.02 | −0.12 |
| HV($C_{22}$) | **0.79*** | 0.12 | 0.01 | −0.26* |
| VV($C_{33}$) | 0.01 | −0.06 | −0.27* | −0.21* |
| Pauli decomposition | | | | |
| HH+VV($T_{11}$) | 0.02 | −0.10 | −0.26* | −0.14 |
| HH-VV ($T_{22}$) | 0.68* | 0.31* | 0.17* | −0.12 |
| Intensity ratio | | | | |
| HH/VV | 0.53* | 0.12 | 0.25* | 0.05 |
| HV/HH | 0.51* | 0.10 | 0.00 | −0.05 |
| HV/VV | **0.92*** | 0.00 | 0.12 | 0.01 |
| Radar vegetation index | | | | |
| RVI | 0.57* | 0.11 | 0.03 | −0.02 |
| Cloude-Pottier decomposition | | | | |
| Entropy($H$) | 0.57* | 0.00 | 0.26* | −0.00 |
| Anisotropy ($A$) | −0.24* | 0.00 | 0.15 | −0.00 |
| Alpha ($\alpha$) | −0.29* | 0.10 | −0.18* | −0.06 |
| Yamaguchi 4-component decomposition | | | | |
| Yamaguchi single bounce | −0.01 | −0.01 | −0.17* | −0.06 |
| Yamaguchi double bounce | 0.43* | 0.14 | 0.08 | −0.01 |
| Yamaguchi volume scattering | 0.68* | 0.17* | 0.00 | −0.03 |
| Yamaguchi helix scattering | 0.35* | 0.01 | 0.00 | −0.12 |

*$p$-Value <0.01.

Moncelloni's study (Macelloni et al. 2001), which showed that the correlation between C-band VV polarization and LAI of broadleaf crops (corn and sunflower) was lower than the correlation between VV polarization and LAI of narrow leaf crops (wheat), and in Xu et al.'s (2014) study, which showed that the classification accuracy of peanut and woods using Radarsat-2 VV polarization was the lowest among the three polarizations (HH, HV, VV).

In Figure 7, the ratio HV/VV is plotted against the corn FVC. The HV/VV gave the best correlation with corn FVC ($R^2$ = 0.92) when the FVC was below 75%. When the corn FVC was greater than 75%, the sensitivity of HV/VV to FVC decreased significantly ($R^2$ = 0.00) (Table 3). In addition to the weakened penetrating capability caused by accumulated

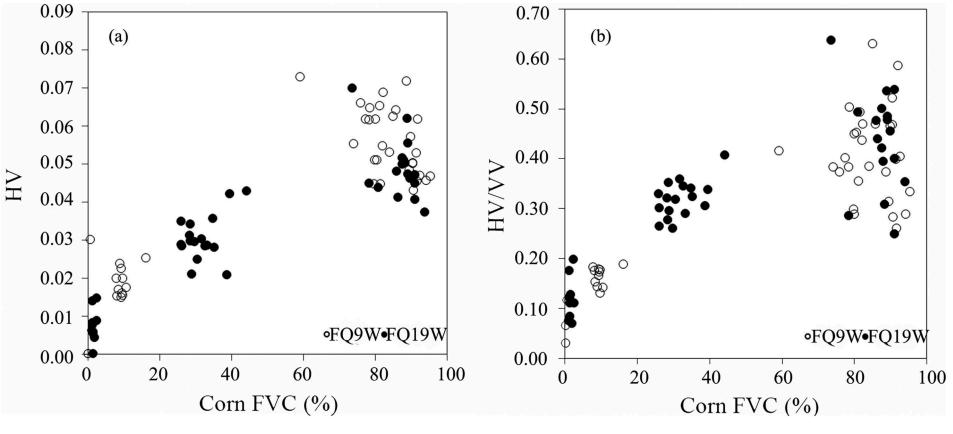



**Figure 7.** Correlation between corn FVC and HV ($C_{22}$) (a) and HV/VV (b).

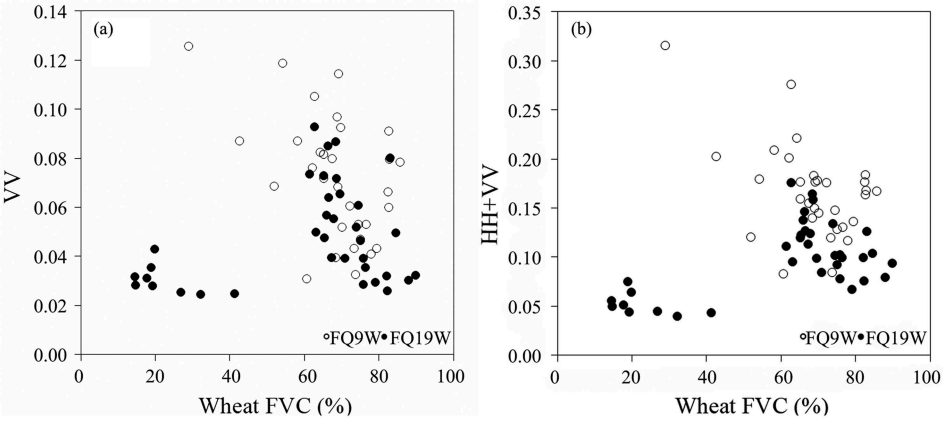



**Figure 8.** Correlation between wheat FVC and VV ($C_{33}$) (a) and HV+VV ($T_{11}$) (b).

biomass, this may also be caused by the faster vertical development of canopy than the horizontal development with the growth of corn and accordingly the saturation of FVC at the later growing stage.

Figure 8 shows the scatterplots between wheat FVC and VV (Figure 8a) and HV+VV (Figure 8b). No obvious correlation was observed between the Radarsat-2 polarimetric SAR signal and the wheat FVC at any stage. It may be because of the two following factors. Firstly, the dense plants and narrow leaves of wheat make the stems contribute a large percentage to the FVC and accordingly lead to the saturation of FVC at an early stage. In addition, the SAR backscatter can easily penetrate the canopy and be affected by the stems and underlying soil due to the upright and narrow characteristics of wheat leaves and the small amount of biomass (Macelloni et al. 2001). In this view, C-band polarimetric SAR parameters are not good indicators for wheat FVC.

The results indicated that the C-band Radarsat-2 polarimetric SAR has a great potential in crop height and FVC estimation for broad-leaf crops (corn), and identifying the changes in crop canopy structures and phenology (corn and wheat) in cloudy areas.

## 4. Conclusions

This study investigated the sensitivity of different Radarsat-2 polarimetric SAR parameters to crop height and FVC of corn and wheat in Southwestern Ontario, Canada. The results showed that the sensitivity of polarimetric SAR parameters to crop variables was higher for corn than for wheat. For corn, the HV, HH-VV, and HV/VV were strongly and positively correlated with the corn height when it was below 150 cm (before the heading stage). The correlation between most of the SAR parameters and corn height became weaker and negative after the heading stage. Only the HH-VV was observed to be strongly correlated with corn height at the later stage. The HV and HV/VV showed good sensitivity to corn FVC when corn FVC was below 75% (before the heading stage). Beyond this threshold, the sensitivity to corn FVC was lost. For wheat, the responses of polarimetric SAR parameters to wheat height were complex. The correlation between polarimetric SAR parameters and wheat height was very weak at the early and later growing stage. And then negative trends

with weak correlations were observed between most polarimetric SAR parameters and wheat height during the stem elongation stage. Among the SAR parameters studied, HV and Yamaguchi helix scattering were negatively correlated with wheat height during this stage with relatively higher coefficients of determination ($R^2 = 0.57$ and $R^2 = 0.39$). Whereas, the wheat FVC was not correlated with any SAR parameters.

## Acknowledgements

This research was supported by NSERC Discovery Grant to Dr. Jinfei Wang. The Radarsat-2 data were provided by Agriculture and Agri-Food Canada (AAFC). The authors would like to acknowledge all the team members from AAFC and GITA lab including Chuiqing Zeng, Qin Ma, and Cheng Qiao, who helped collecting the field data.

## Disclosure statement

No potential conflict of interest was reported by the authors.

## Funding

This research was supported by the NSERC Discovery Grant to Dr. Jinfei Wang.

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